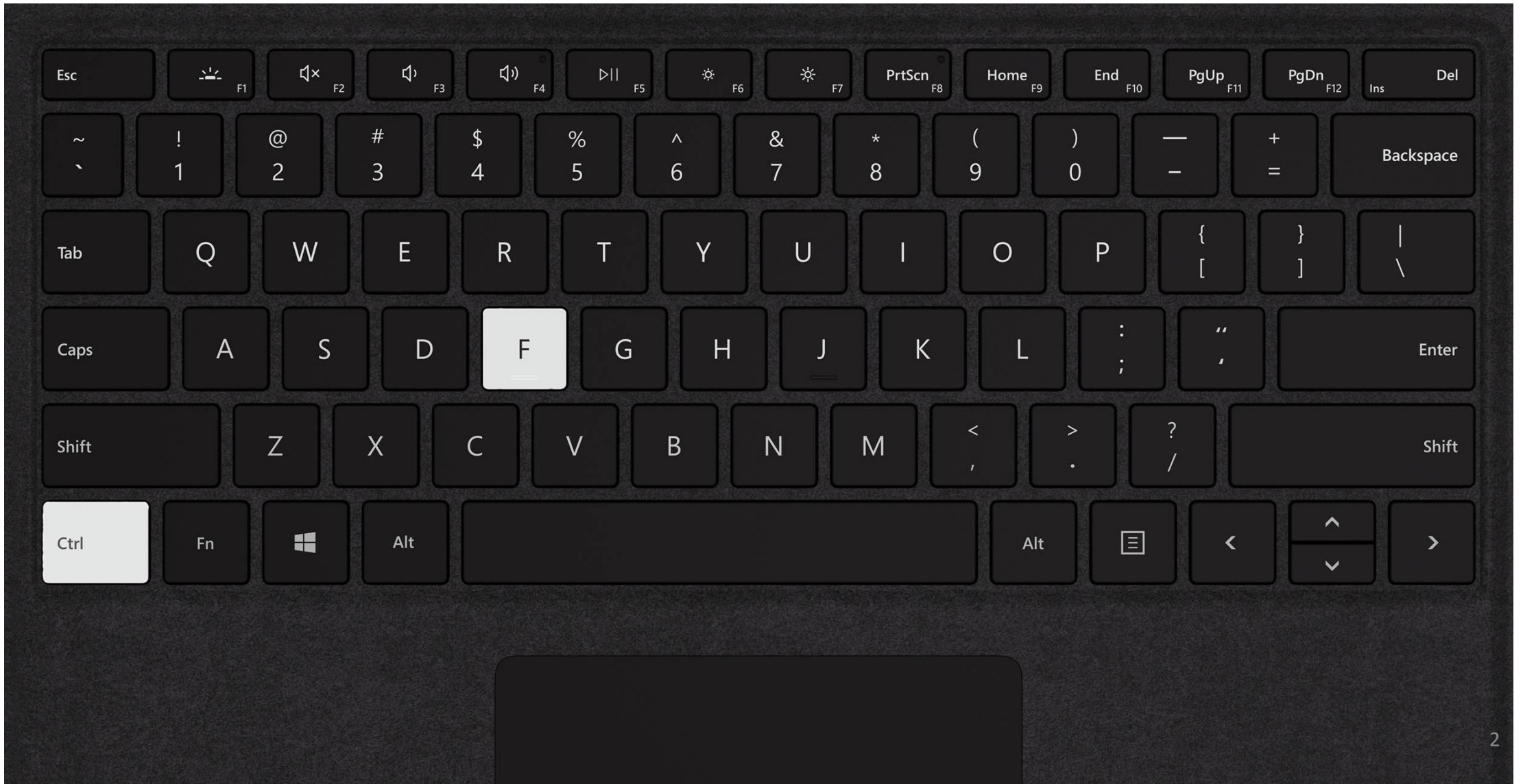


Research project: *Echoes from the Past:*
Unveiling a Lost Soundscape with Digital Analysis
<https://doi.org/10.54499/2022.01957.PTDC>

THE MUSIC ENCODING INITIATIVE

ENCODING OF EARLY MUSIC

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et crucem subire fecisti. concede
 ꝑꝑici ut et paciencie ꝑꝑsi habere
 documta. et resurrectionis ei con
 sortia mereamur. *Ps. ad philipenses.*
Frs: *h*oc sentite
 in uob: qđ et in xpo ihu. qđ
 cū in forma dei ēēt nō sa
 pīā arbitrat' est se se equalē
 deo: sed semet ꝑꝑsi exinaniuit
 formā seruū accipien'. in similitu
 dinē hominū factus. et habitu
 inuenit' ut homo. humiliavit
 semet ꝑꝑsi factus obediens usq;
 ad mortē: mortē autē crucis. Et
 qđ et d's exaltauit illū. et donauit
 illi nom' qđ ē sup omē nom': ut in
 nomine ihu omē genua flectat'.
 celestium terrestrium et infernoz.
 Et omis lingua confiteatur: quia
 dñs ihu xpc in glā ē dei patris.

o enuth ma nuni deverti in me a
in volunrate in a de davi th me
er cum gloria a sum pti th me
uam bonis a spali de u f
re a co rde mei auren
a me ri furi ve de f per

1/33



ribi regnanti. p[ro]p[ri]um. ecc[lesi]e. m[ag]is. **G**loria laus
 lebi. sancte. i[esu]s. uenit. sancta. i[esu]s. uenit.
 cum uero. i[esu]s. carmina. digna. sepe. **G**loria
 laus. **R**egrediente dno
Mentem familie tue. **O**
 q[ui]s d[omi]ne intercedente beato. al.
 i munere cōpunctionis ap[er]it lar
 gitate pietatis exaudi. **P** admiss
 iam. **O**ffm
ONNE
NRO
FALAS
 a. v. l. i. q. r. e. x. m. a. m. e. a. o. d. e. f. e. r. e. t. o.
 n. e. m. e. a. m. a. s. p. i. c. e. l. i. b. e. r. a. m. e. d. e. o. r. e. l. e. o. n. g.
 i. a. c. o. r. d. i. b. u. s. u. n. i. c. o. r. d. i. u. m. h. u. m. i. l. i. t. a. t. e. m. e.
 a. m. **P** d. e. u. s. d. e. u. s. i. s. t. a. s. e. u. a. u. a. e. **O**
O m[ne]s sēp[er]t[em]p[or]e d[omi]n[us] qui
 hūmāno generi ad imitā
 dū hūmilitatis exemplū
 saluātem n[ost]rā carnem sūm[us]

et crucem subne fecisti. concede
 p[ro]p[ri]i ut et patientie ip[s]i habere
 docum[en]ta. et resurrectionis ei[us] con
 sortia meream[ur]. **P** ad philipenses.
Rs. **H**oc sentite
F in uob[is]. q[uod] et in x[p]o ihu. q[ui]
 cū in forma dei ēēt nō ra
 pinā arbitrat[us] est. ēē se equalē
 deo. sed semet ip[s]ū exinaniuit
 formā serui accipien[tes]. in similitu
 dinē hominū factus. et habitu
 inuenit[ur] ut homo. humiliavit
 semet ip[s]ū factus obediens usq[ue]
 ad mortē. mortē aut[em] crucis. q[ui]
 q[uod] et d[omi]n[us] exaltauit illū. et donauit
 illi nom[en] q[uod] ē sup[er] om[n]e nom[en]. ut in
 nom[in]e ihu om[n]e genu flectat[ur].
 celestium terestrium et inferno[rum].
 et om[n]is lingua confiteatur. quia
 d[omi]n[us] ihu x[p]c in gl[ori]a ē dei patris.

Constiti. ma. num. de. veni. m. me. a.
 m. uoluntate. m. a. de. d. u. m. m. e.
 er. cum. glori. a. s. h. u. m. i. l. i. t. a. t. e. m. e.
 uam. bonis. affat. de. u. s.
 re. ch. co. r. d. e. m. e. a. u. t. e. n. t. i. a.
 a. m. e. m. o. r. i. s. u. n. t. p. e. n. e.

quilisma

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×



Optical Music Recognition (OMR) is the transformation of images of music notation into digital representations with limited direct human involvement.

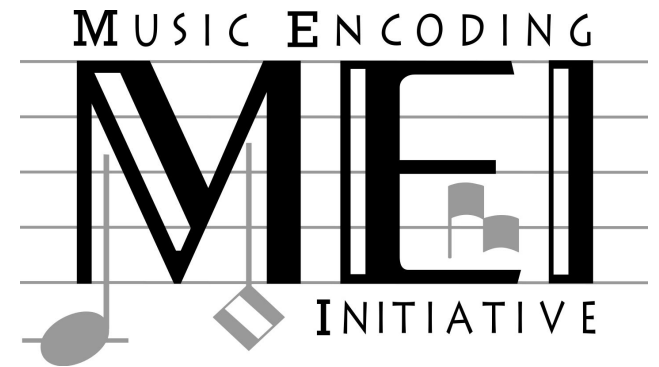


Why Would We Want to Encode Music

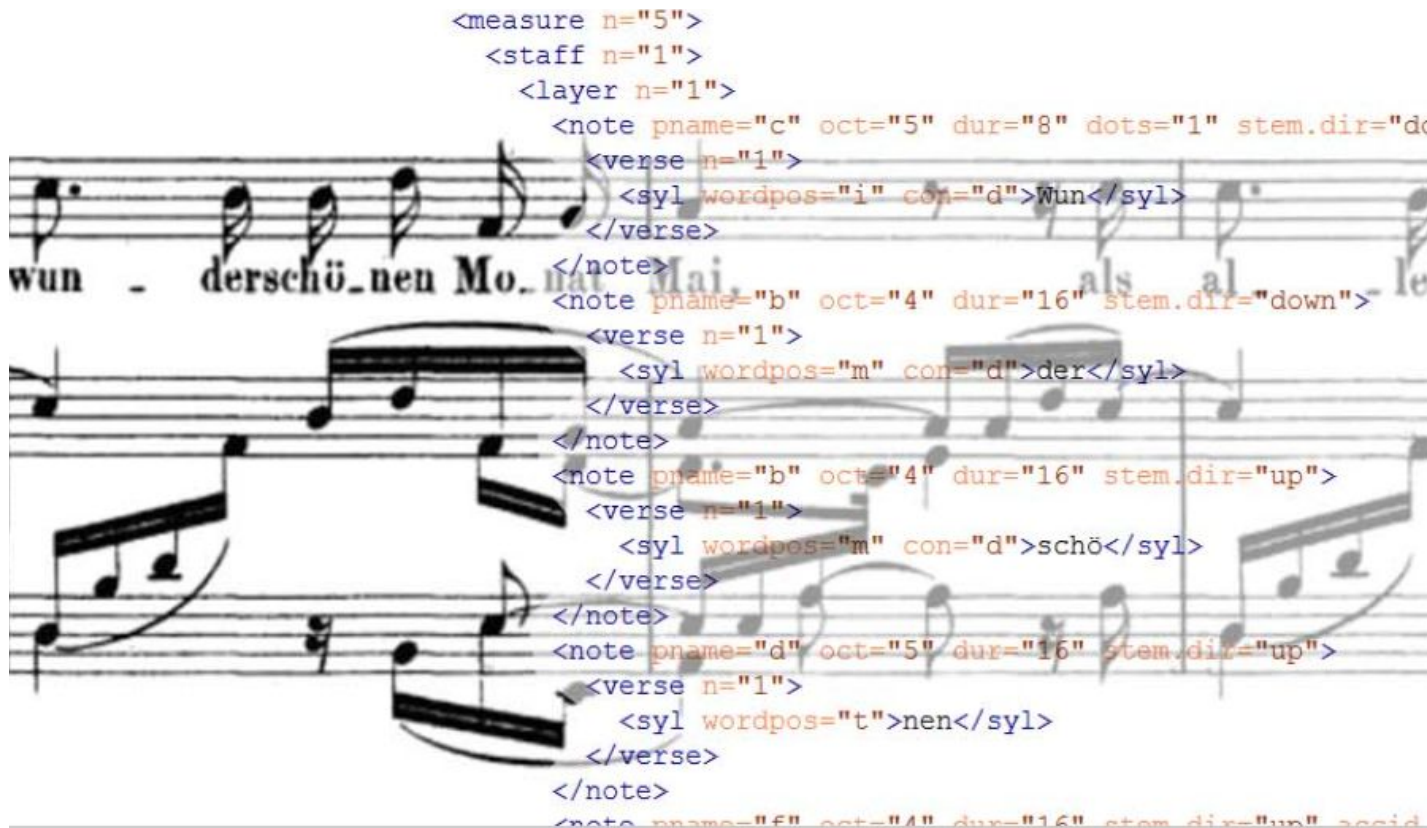
- Machine-readable files → ask the computer:
 - Perform searches
 - Melodic search
 - Search for ornamental neume shapes
 - Look for melismatic passages
 - Music analysis
 - Cross-comparison
 - Compute the mode based on: finalis, repercussio, and range

Why Would We Want to Encode Music in **MEI**?

The Music Encoding Initiative (**MEI**), inspired by the Text Encoding Initiative (**TEI**), is a joint effort of an open and cosmopolitan community of technologists, musicologists, musicians, librarians, and music enthusiasts who discuss and approve rules aiming at fully capturing music documents into a machine-readable format

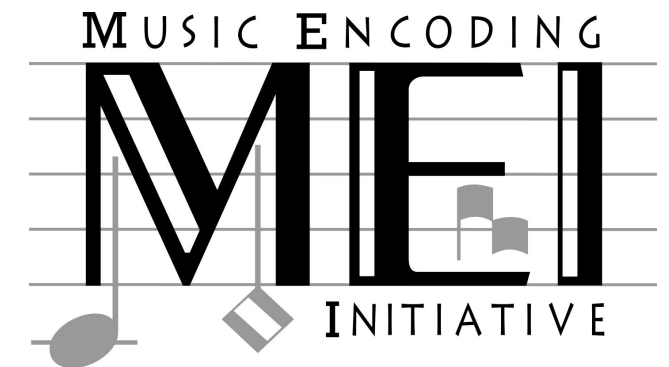


Why Would We Want to Encode Music in MEI?



The image shows a musical score with lyrics and MEI code overlaid. The lyrics are "wun - der schö - nen Mo - nat Mai, als al - le". The MEI code is as follows:

```
<measure n="5">
  <staff n="1">
    <layer n="1">
      <note pname="c" oct="5" dur="8" dots="1" stem.dir="down">
        <verse n="1">
          <syl wordpos="i" con="d">Wun</syl>
        </verse>
      </note>
      <note pname="b" oct="4" dur="16" stem.dir="down">
        <verse n="1">
          <syl wordpos="m" con="d">der</syl>
        </verse>
      </note>
      <note pname="b" oct="4" dur="16" stem.dir="up">
        <verse n="1">
          <syl wordpos="m" con="d">schö</syl>
        </verse>
      </note>
      <note pname="d" oct="5" dur="16" stem.dir="up">
        <verse n="1">
          <syl wordpos="t">nen</syl>
        </verse>
      </note>
      <note pname="f" oct="4" dur="16" stem.dir="up" accid="f">
```



OMR recognizes the symbols on the page

MEI provides information on the *meaning* of the musical symbols

Why Would We Want to Encode Music in **MEI**?

MEI (or *Music Encoding Initiative*) is a machine-readable format that allows for the encoding of **early music** among other things:

- **Neume | Mensural | Tablature** (plus common Western music notation, CMN)
- **Rich metadata**
- **Relation to facsimiles & recordings**
- **Markup for Analysis and Harmonies**
- **Scholar editions**

Why Would We Want to Encode Music in MEI?

MEI (or *Music Encoding Initiative*) is a machine-readable format that allows for the encoding of **early music** among other things:

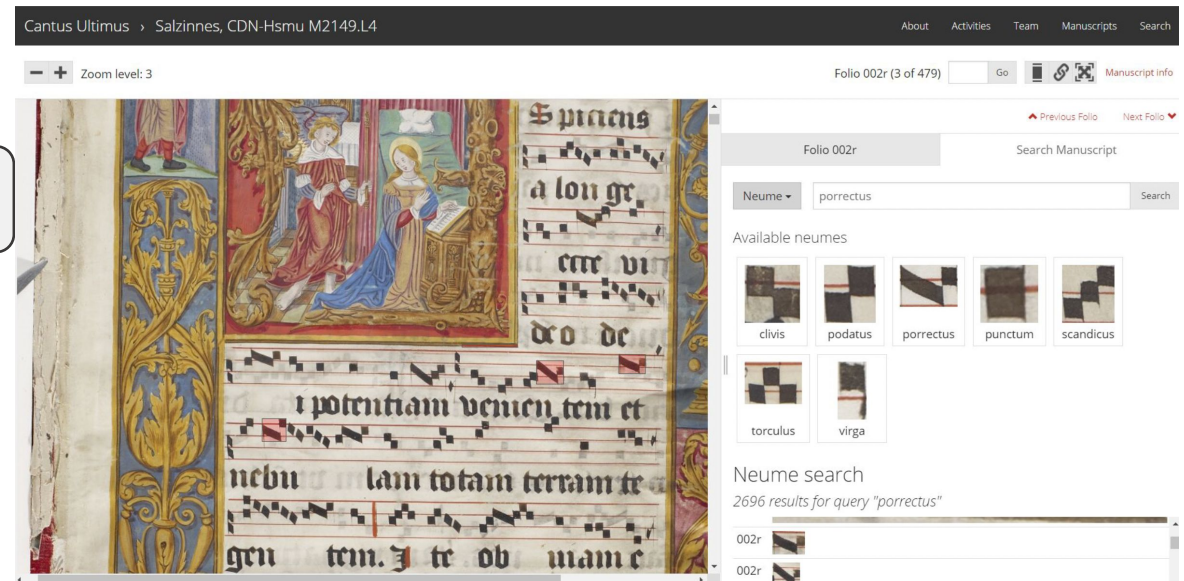
- **Neume | Mensural | Tablature** (plus common Western music notation, CMN)

- **Rich metadata**

- **Relation to facsimiles & recordings**

- **Markup for Analysis and Harmonies**

- **Scholar editions**

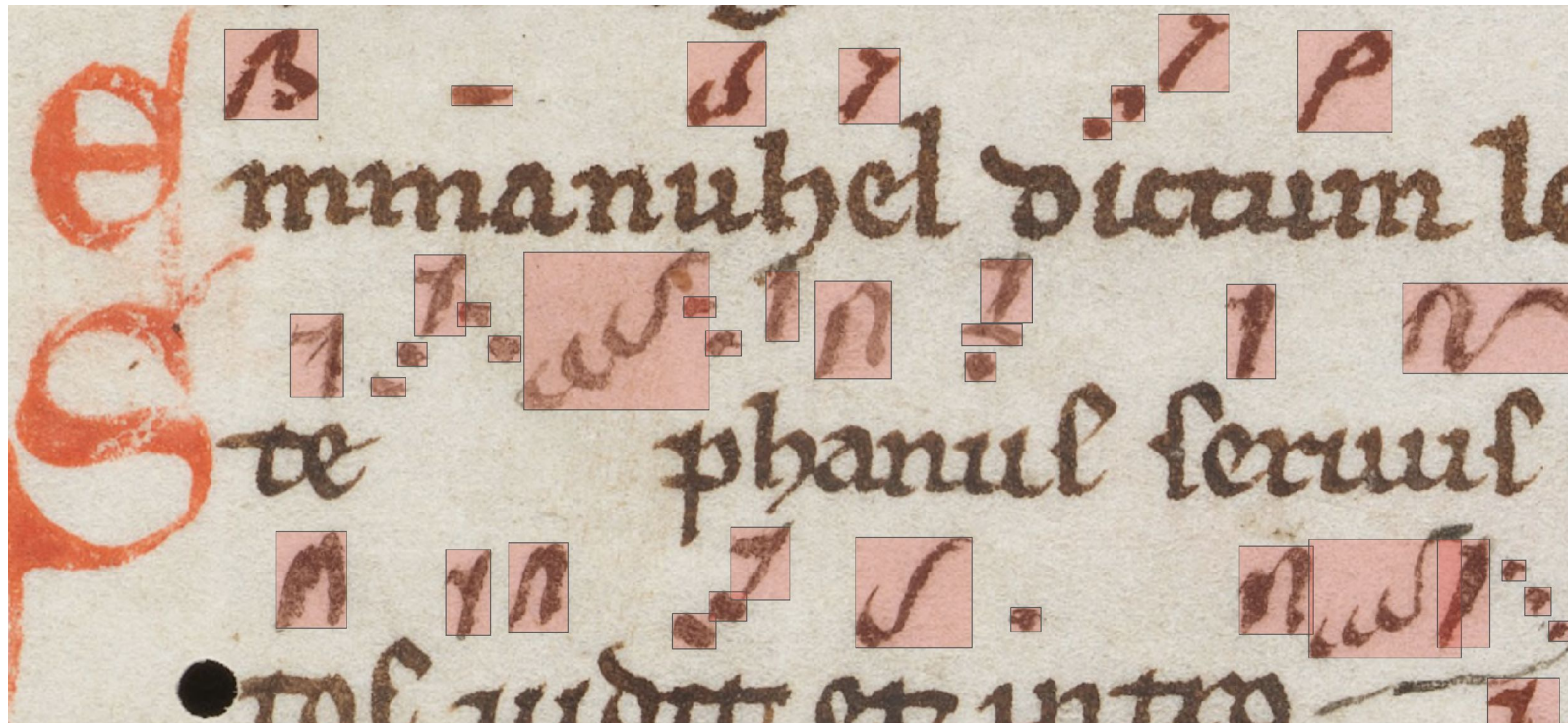


Two practical examples #1



```
<neume>  
  <nc curve= "a"/>  
  <nc tilt="ne" intm="d"/>  
</ neume >
```


Two practical examples #1



Two practical examples #2

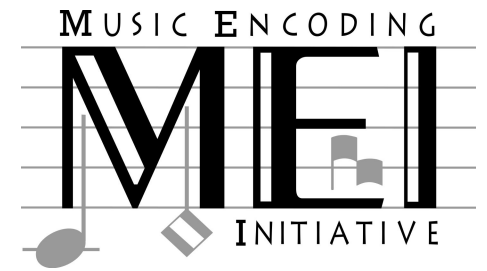


```
<neume>  
  <nc tilt="ne"/>  
  <nc tilt="se" relen="s" intm="d"/>  
</neume>
```

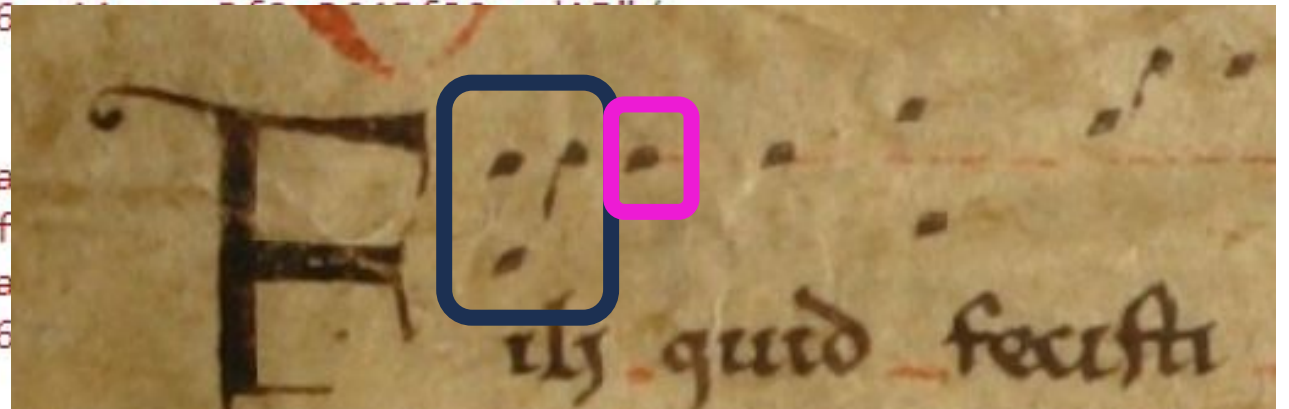


```
<neume>  
  <nc tilt="ne"/>  
  <nc tilt="se" relen="l" intm="d"/>  
</neume>
```

MEI is a hierarchical **system of tags** employed to describe in a machine-readable language the notation of a specific manuscript



```
<syllable xml:id="m-43558eee-8a6a-11ee-a3f2-3645f29ecd15">
  <syl xml:id="m-43558f20-8a6a-11ee-a3f2-3645f29ecd15">Fi</syl>
  <neume xml:id="m-43558f52-8a6a-11ee-a3f2-3645f29ecd15">
    <nc loc="0" xml:id="m-43558f8e-8a6a-11ee-a3f2-3645f29ecd15"/>
    <nc loc="-2" xml:id="m-43558fc0-8a6a-11ee-a3f2-3645f29ecd15"/>
    <nc loc="0" tilt="ne" xml:id="m-43558ff2-8a6a-11ee-a3f2-3645f29ecd15"/>
  </neume>
</syllable>
<syllable xml:id="m-43559024-8a6a-11ee-a3f2-3645f29ecd15">
  <syl xml:id="m-4355904c-8a6a-11ee-a3f2-3645f29ecd15">li</syl>
  <neume xml:id="m-4355907e-8a6a-11ee-a3f2-3645f29ecd15">
    <nc loc="0" xml:id="m-435590b0-8a6a-11ee-a3f2-3645f29ecd15"/>
  </neume>
</syllable>
<syllable xml:id="m-435590d8-8a6a-11ee-a3f2-3645f29ecd15">
  <syl xml:id="m-4355910a-8a6a-11ee-a3f2-3645f29ecd15">
  <neume xml:id="m-4355913c-8a6a-11ee-a3f2-3645f29ecd15">
    <nc loc="0" xml:id="m-43559178-8a6a-11ee-a3f2-3645f29ecd15"/>
  </neume>
</syllable>
```



Short Introduction to MEI



Music Encoding Initiative (MEI)

- A community and a format
- **The format:** Encodes music notation in XML (*eXtensible Markup Language*)
 - Hierarchical encoding (tree structure)
 - Its core objects are **elements** (tags, e.g. `<note>`)
 - Opening tags & closing tags → when there is content in between
 - Self-closing tags → when there is no content to be included in between tags
 - Uses **attributes** to define the properties of elements
 - XML does not "do" anything, it is just information wrapped in tags
- **Goal:** Encode a wide variety of music documents

	<code><beam></code> Opening tag <code><note pname="f" oct="4" dur="8"/></code> Self-closing tag <code><note pname="c" oct="4" dur="8"/></code> <code></beam></code> Closing tag
--	--

XML is a hierarchical encoding system



```
<neume>  
  <nc tilt="e" />  
  <nc tilt="n" intm="u" />  
</neume>
```

<neume> is the parent element = Sign representing one or more musical pitches

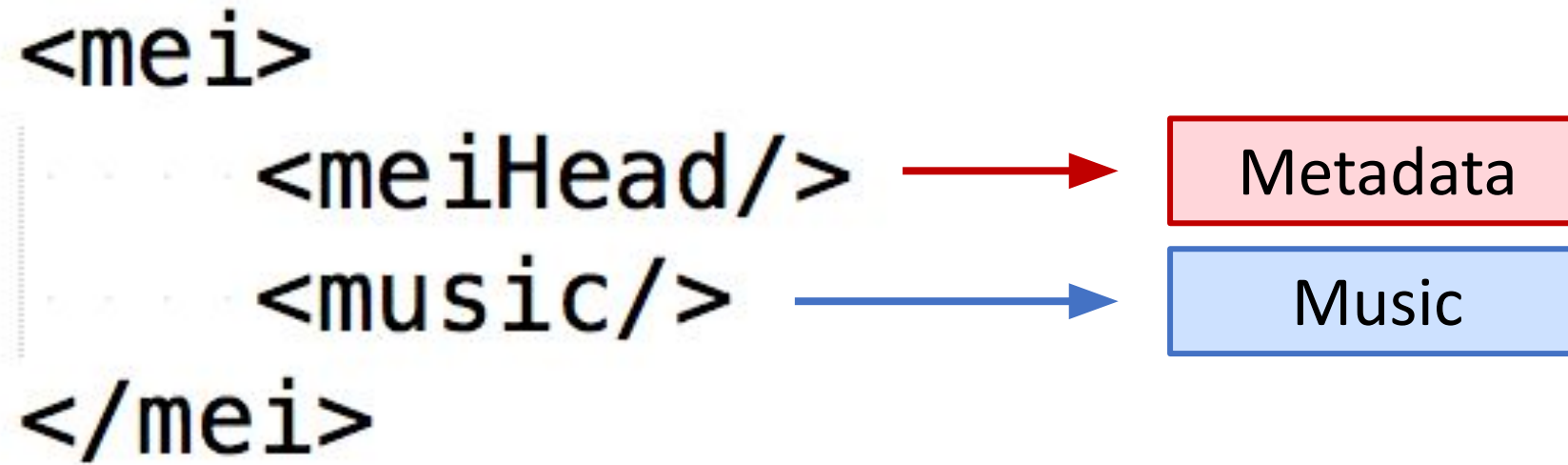
<nc> is a child element = Sign representing a single pitched event, although the exact pitch may not be known

@tilt is an attribute of **<nc>** = Direction of the pen-stroke

@intm is an attribute of **<nc>** = Encodes the melodic interval from the previous pitch

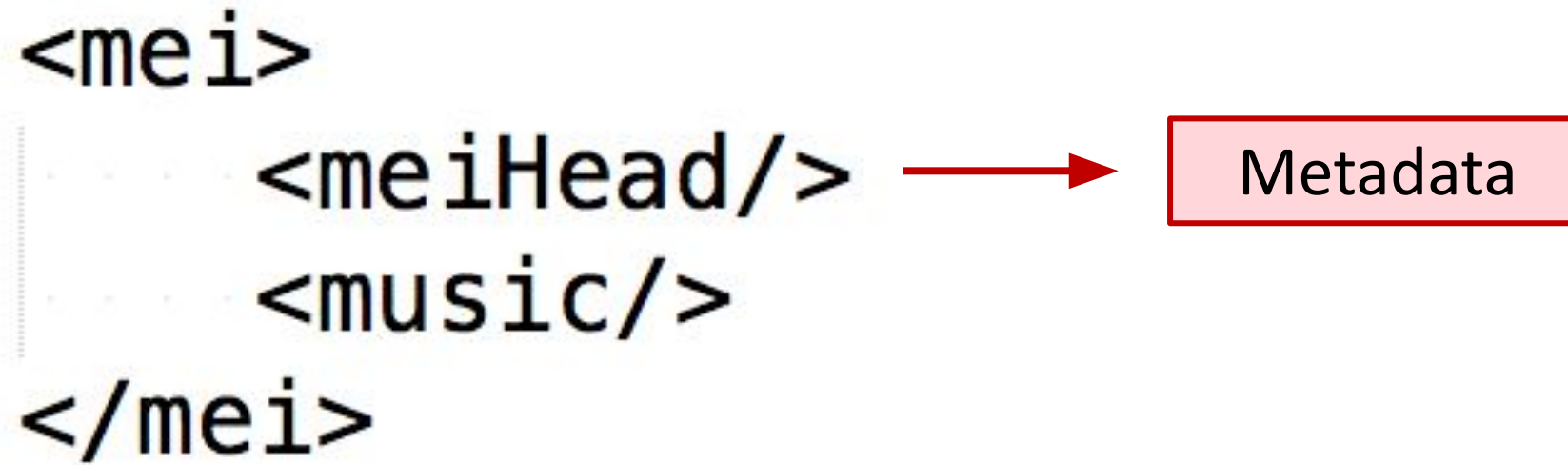
Basic Structure of an MEI File

Basic Structure of an MEI File



Basic Structure of an MEI File

```
<mei>  
  <meiHead/> → Metadata  
  <music/>  
</mei>
```



<meiHead> basic elements

```
<mei>
  <meiHead>
    <fileDesc>
      <titleStmt>
        <title></title>
      </titleStmt>
      <pubStmt/>
    </fileDesc>
  </meiHead>
  <music/>
</mei>
```

TUTORIALS

[https://music-encoding.org/
resources/tutorials.html](https://music-encoding.org/resources/tutorials.html)

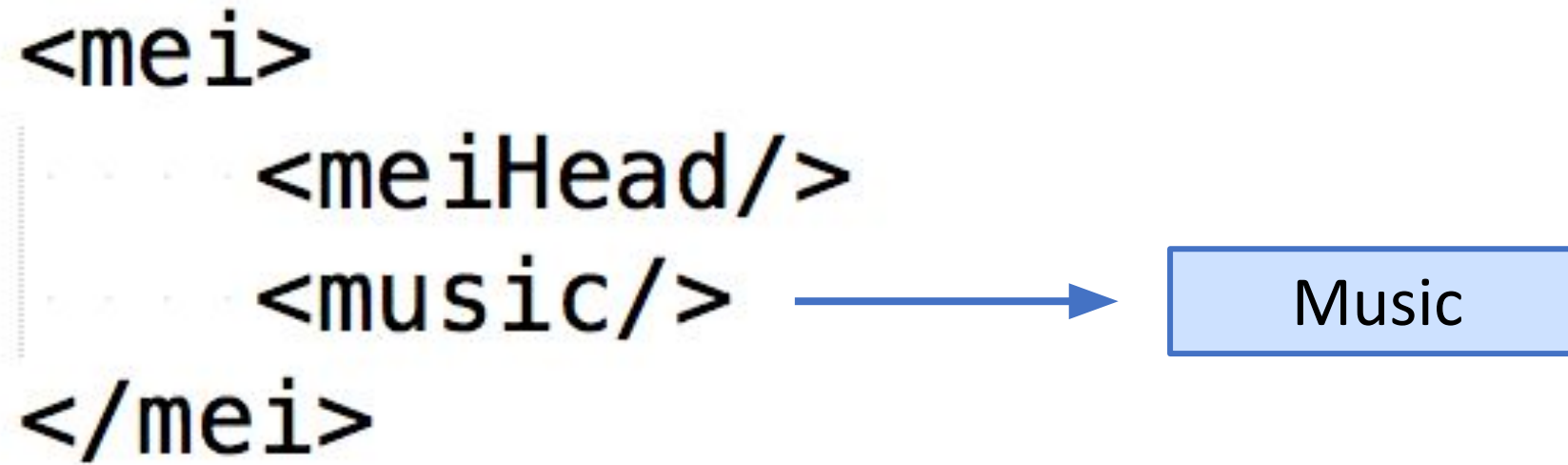
XML Basics and Minimal MEI File Structure

Outermost basic structure of
an **MEI-conformant** document
(conforms to the **schema** of MEI)

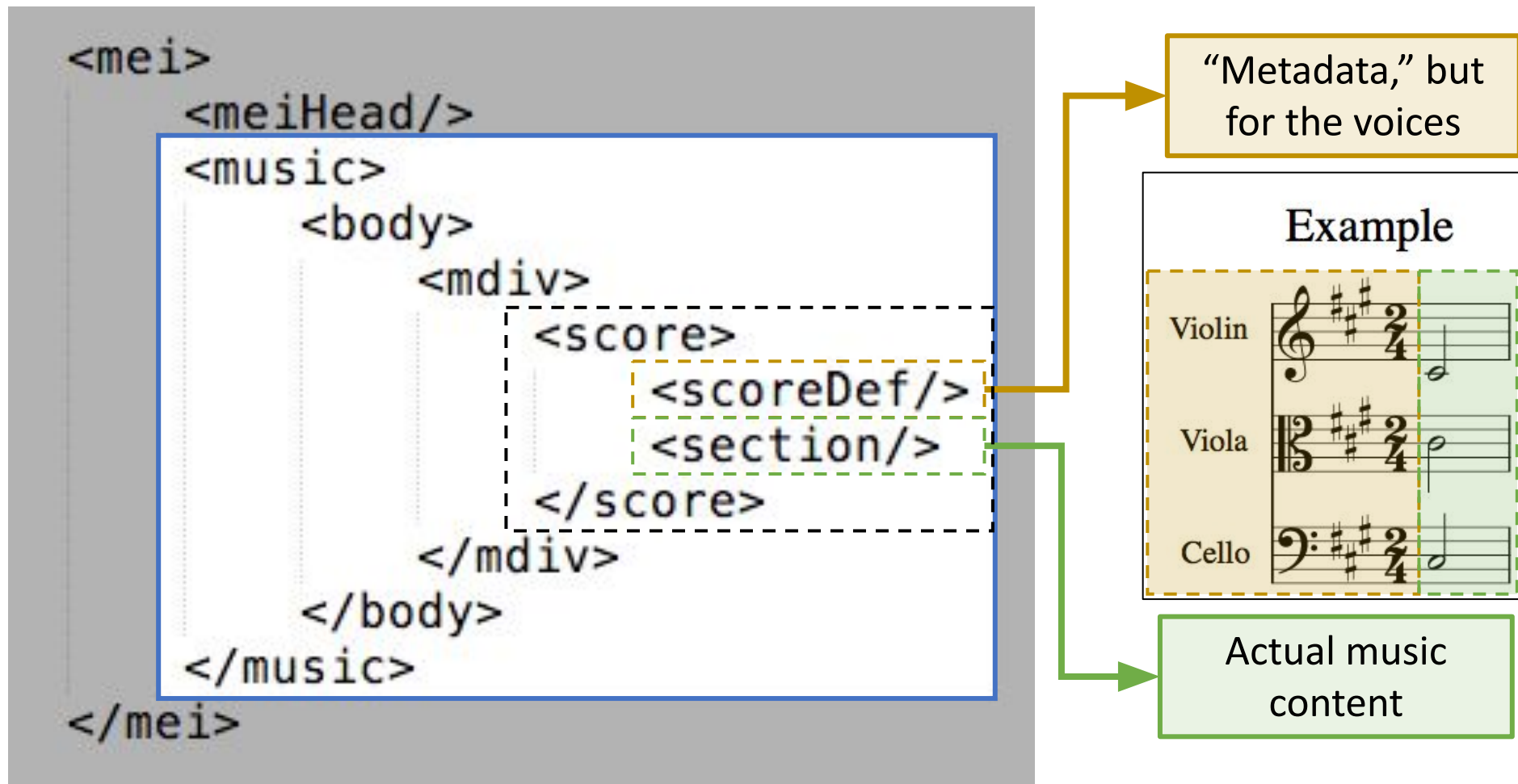
Basic structure
of a “valid” MEI file

Basic Structure of an MEI File

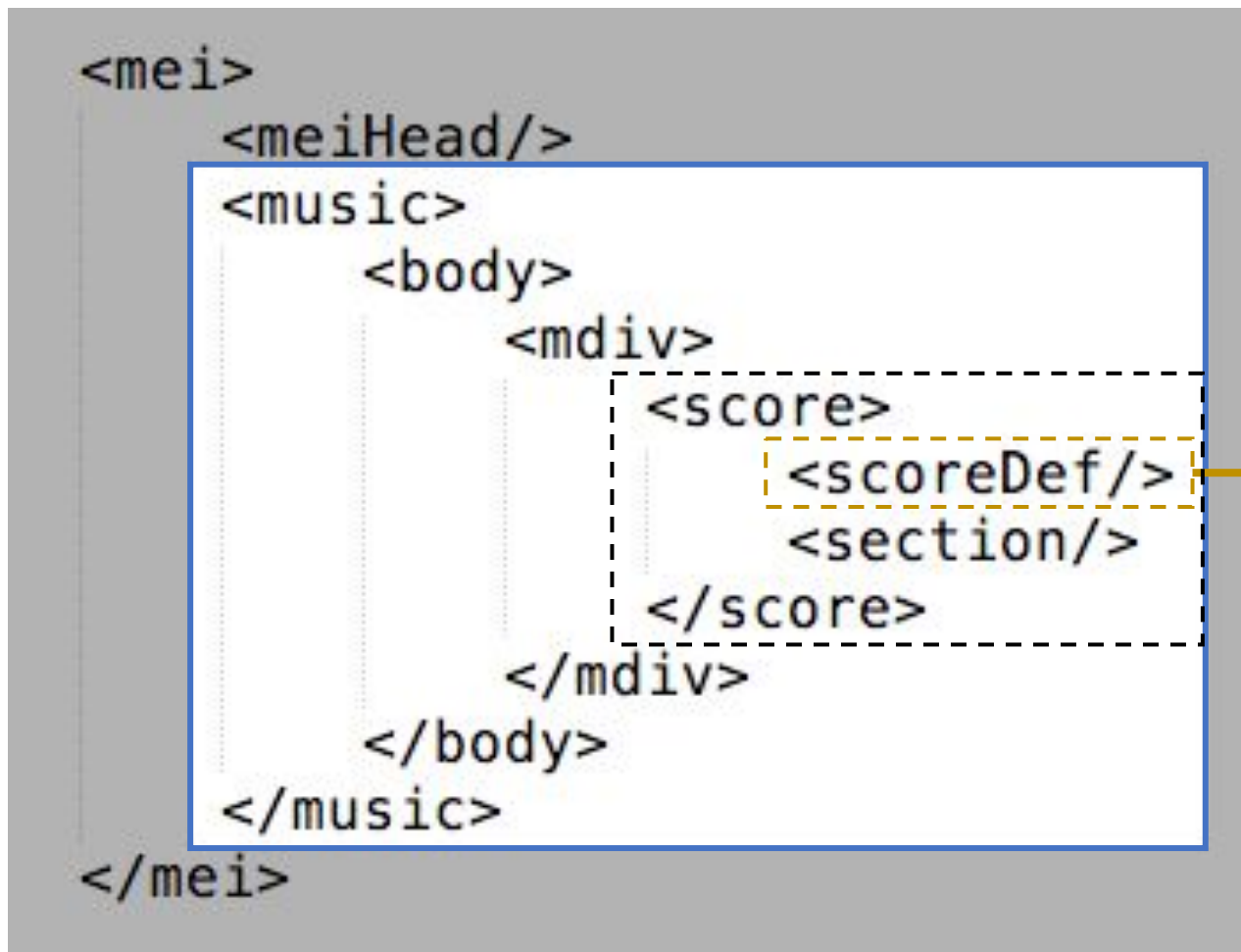
```
<mei>  
  <meiHead/>  
  <music/> → Music  
</mei>
```

The diagram illustrates the basic structure of an MEI file. It shows the XML code: `<mei>`, `<meiHead/>`, `<music/>`, and `</mei>`. A blue arrow points from the `<music/>` tag to a light blue rectangular box labeled "Music".

<music> basic elements

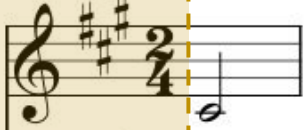
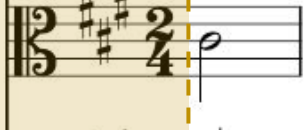
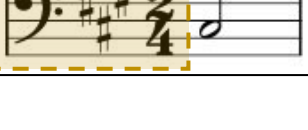


<music> basic elements



“Metadata,” but
for the voices

Example

Violin	
Viola	
Cello	

The example shows three staves of musical notation for Violin, Viola, and Cello. Each staff has a key signature of two sharps (F# and C#) and a 2/4 time signature. The Violin staff uses a treble clef, the Viola staff uses an alto clef, and the Cello staff uses a bass clef. Each staff contains a single quarter note. A dashed yellow box highlights the entire example section.

<scoreDef> - General Information for Voices

```
<score>
  <scoreDef>
    <staffGrp>
      <staffDef n="1"/>
      <staffDef n="2"/>
      <staffDef n="3"/>
    </staffGrp>
  </scoreDef>
  <section/>
</score>
```

General information for **ALL** voices

- Meter
- Key

Specific information for **EACH** voice

- Clef
- Label
- Number of staff-lines (@lines)
- @n

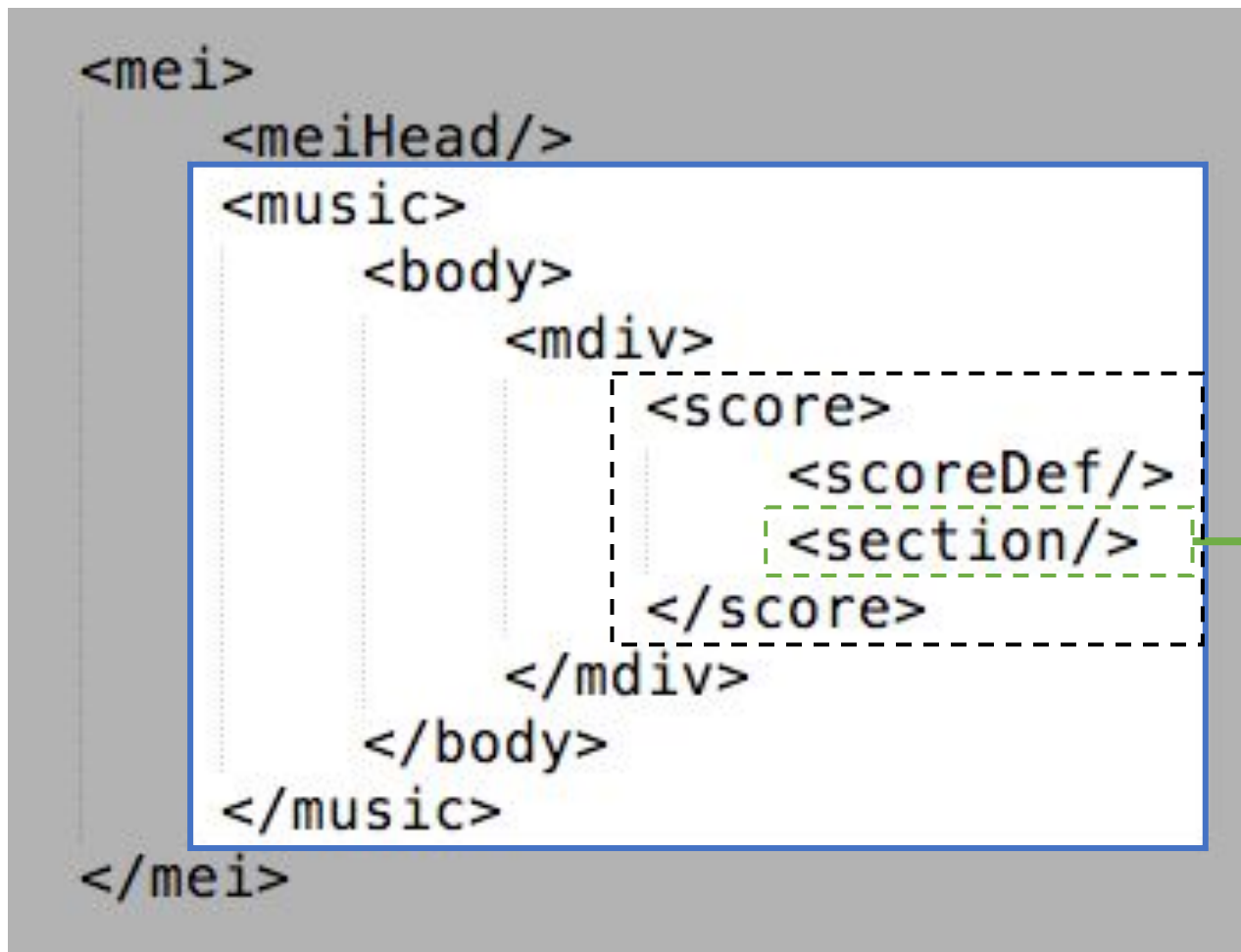
required for validation

Example

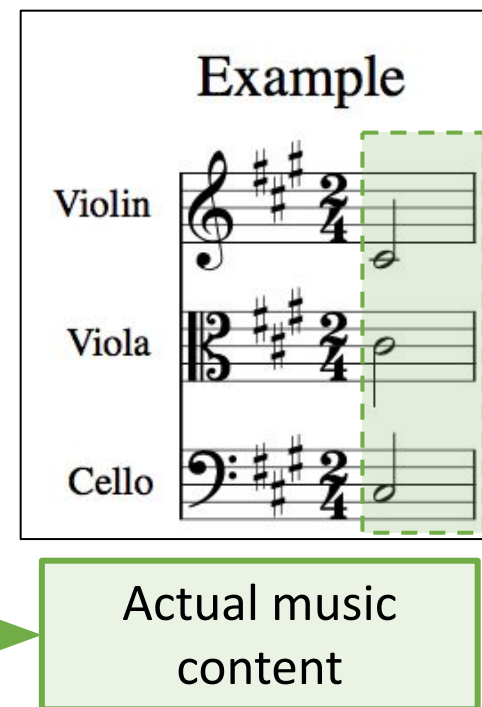
The example shows three staves of music. The top staff is labeled 'Violin' and uses a treble clef. The middle staff is labeled 'Viola' and uses an alto clef. The bottom staff is labeled 'Cello' and uses a bass clef. All three staves are in the key of D major (two sharps) and 2/4 time. The first measure of each staff contains a half note. The staves are grouped together in a dashed box, and the labels are in dashed ovals.

<https://music-encoding.org/guidelines/v5/content/shared.html>

<music> basic elements



Example

A musical notation example showing three staves: Violin, Viola, and Cello. The key signature is two sharps (F# and C#) and the time signature is 2/4. The Violin staff has a treble clef, the Viola staff has an alto clef, and the Cello staff has a bass clef. Each staff contains a single note. A green dashed box highlights the notes on all three staves. A green arrow points from the <section/> element in the XML diagram to this box.

Violin

Viola

Cello

Actual music content

<section> - Actual Music

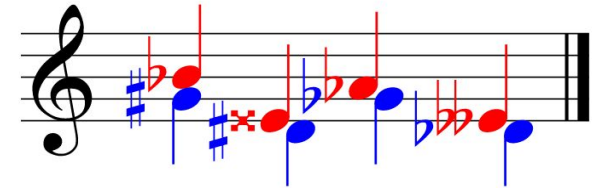
```
<score>
  <scoreDef/>
  <section>
    <measure n="1">
      <staff n="1">
        <layer>
          <!-- MUSIC WITHIN THE STAFF -->
        </layer>
      </staff>
      <staff n="2"/>
      <staff n="3"/>
    </measure>
    ...
    <measure n="20"/>
  </section>
</score>
```

<https://music-encoding.org/guidelines/v5/content/shared.html>

<section> - Actual Music

```
<score>
  <scoreDef/>
  <section>
    <measure n="1">
      <staff n="1">
        <layer>
          <!-- MUSIC WITHIN THE STAFF -->
        </layer>
      </staff>
      <staff n="2"/>
      <staff n="3"/>
    </measure>
    ...
    <measure n="20"/>
  </section>
</score>
```

<layer n="1">



<layer n="2">

<https://music-encoding.org/guidelines/v5/content/shared.html>

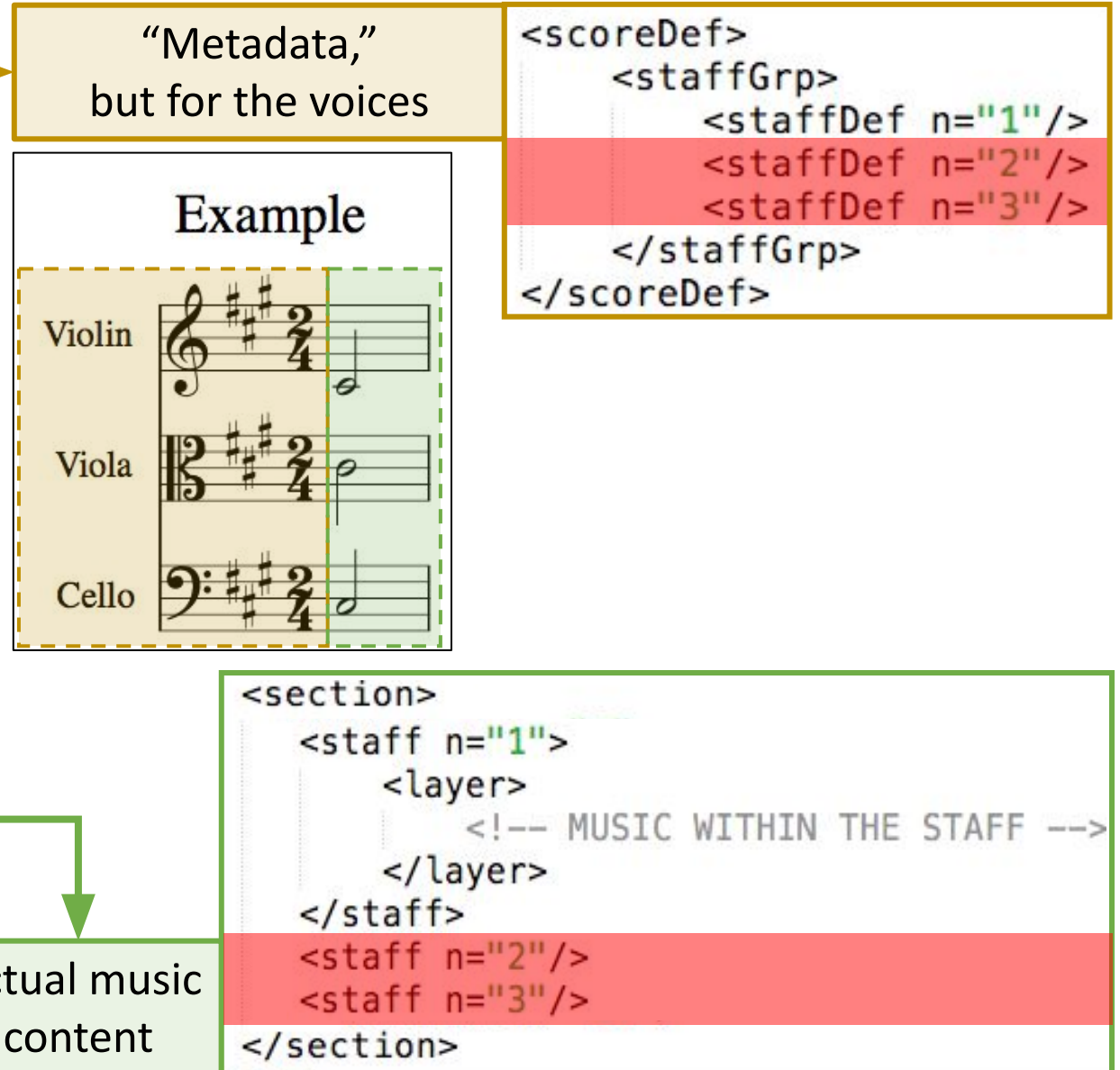
Early Music has No Measures!

```
<score>
  <scoreDef/>
  <section>
    <measure n="1">
      <staff n="1">
        <layer>
          <!-- MUSIC WITHIN THE STAFF -->
        </layer>
      </staff>
      <staff n="2"/>
      <staff n="3"/>
    </measure>
    ...
    <measure n="20"/>
  </section>
</score>
```


Monophonic Music has a Single Staff

```
<score>
  <scoreDef/>
  <section>
    <staff n="1">
      <layer>
        <!-- MUSIC WITHIN THE STAFF -->
      </layer>
    </staff>
    <staff n="2"/>
    <staff n="3"/>
  </section>
</score>
```

Common Structure for Monophonic Music



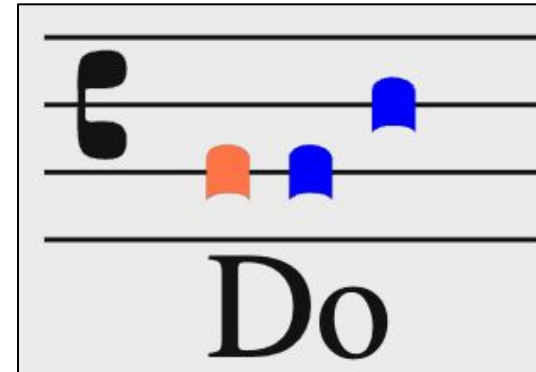
MEI for Neume Notation

Most neume notation is used to set music to an existing text. Therefore, the “syllable” is the fundamental unit of structure, with the neumes themselves serving as a means of “sonifying” the text.

Therefore, the syllable element provides high-level organization in this repertoire.

Elements

- `<syllable>`
 - **One** `<syl>` child
 - **One or more** `<neume>` children
 - neume component `<nc>` children, these are the individual notes that form part of the neume



```
<syllable>  
  <syl>Do</syl>  
  <neume>  
    <nc loc="2" />  
  </neume>  
  <neume>  
    <nc loc="2" />  
    <nc loc="4" />  
  </neume>  
</syllable>
```

Elements

- <syllable>
 - **One** <syl> child
 - **One or more** <neume> children
 - neume component <nc> children, these are the individual notes that form part of the neume

Attributes of neume components:

○ Pitch

- ❖ @pname = c | d | e | f | g | a | b
- ❖ @oct = ... 2 | 3 | 4 ...
- ❖ @loc: position in the staff (integer, 0 is the first line) = ... -2 | -1 | 0 | 1 | 2 ...
- ❖ @intm: melodic interval (integer: negative, zero, positive) = ... -2 | -1 | 0 | 1 | 2 ...

○ Shape

- ❖ @tilt: direction of penstroke = n | s | se | ...
- ❖ @curve = c | a
- ❖ ...

Elements

- <syllable>
 - One <syl> child
 - One or more <neume> children
 - neume component <nc> children, these are the individual notes that form part of the neume

Attributes of neume components:

○ Pitch

- ❖ @pname = c | d | e | f | g | a | b
- ❖ @oct = ... 2 | 3 | 4 ...
- ❖ @loc: position in the staff (integer, 0 is the first line) = ... -2 | -1 | 0 | 1 | 2 ...
- ❖ @intm: melodic interval (integer: negative, zero, positive) = ... -2 | -1 | 0 | 1 | 2 ...

○ Shape

- ❖ @tilt: direction of penstroke = n | s | se | ...



```
<neume>
  <nc pname="d" oct="3" tilt="n" />
  <nc pname="c" oct="3" />
</neume>
```


Elements

- `<syllable>`
 - One `<syl>` child
 - One or more `<neume>` children
 - neume component `<nc>` children, these are the individual notes that form part of the neume

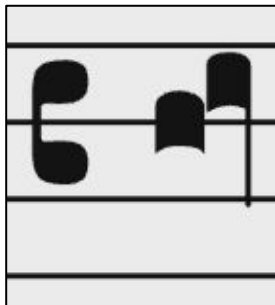
Attributes of neume components:

○ Pitch

- ❖ `@pname` = c | d | e | f | g | a | b
- ❖ `@oct` = ... 2 | 3 | 4 ...
- ❖ `@loc`: position in the staff (integer, 0 is the first line) = ... -2 | -1 | 0 | 1 | 2 ...
- ❖ `@intm`: melodic interval (integer: negative, zero, positive) = ... -2 | -1 | 0 | 1 | 2 ...

○ Shape

- ❖ `@tilt`: direction of penstroke = n | s | se | ...
- ❖ `@curve` = c | a
- ❖ ...



```
<neume>  
  <nc loc="4" />  
  <nc intm="1" tilt="s" />  
</neume>
```

Elements

- `<syllable>`
 - **One** `<syl>` child
 - **One or more** `<neume>` children
 - neume component `<nc>` children, these are the individual notes that form part of the neume

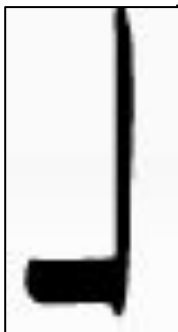
Attributes of neume components:

○ Pitch

- ❖ `@pname` = c | d | e | f | g | a | b
- ❖ `@oct` = ... 2 | 3 | 4 ...
- ❖ `@loc`: position in the staff (integer, 0 is the first line) = ... -2 | -1 | 0 | 1 | 2 ...
- ❖ `@intm`: melodic interval (integer: negative, zero, positive) = ... -2 | -1 | 0 | 1 | 2 ...

○ Shape

- ❖ `@tilt`: direction of penstroke = n | s | se | ...
- ❖ `@curve` = c | a
- ❖ ...



```
<neume>
  <nc tilt="e" />
  <nc tilt="n" intm="u" />
</neume>
```

Elements

- <syllable>
 - One <syl> child
 - One or more <neume> children
 - neume component <nc> children, these are the individual notes that form part of the neume



```
<neume>
  <nc tilt="ne" />
  <nc tilt="se" curve="c"
    intm="d" />
</neume>
```

Old hispanic notation

Attributes of neume components:

○ Pitch

- ❖ @pname = c | d | e | f | g | a | b
- ❖ @oct = ... 2 | 3 | 4 ...
- ❖ @loc: position in the staff (integer, 0 is the first line) = ... -2 | -1 | 0 | 1 | 2 ...
- ❖ @intm: melodic interval (integer: negative, zero, positive) = ... -2 | -1 | 0 | 1 | 2 ...

○ Shape

- ❖ @tilt: direction of penstroke = n | s | se | ...
- ❖ @curve = c | a
- ❖ ...

Elements

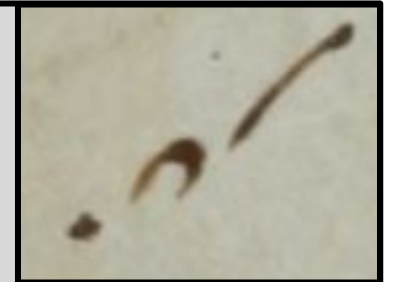
- **<syllable>**
 - **One** **<syl>** child
 - **One or more** **<neume>** children
 - neume component **<nc>** children, these are the individual notes that form part of the neume
 - **<nc>** can have children too:
 - **<episema>** ○ **<oriscus>**
 - **<hispanTick>** ○ **<quilisma>**
 - **<liquescent>** ○ **<strophicus>**

https://music-encoding.org/guidelines/v5/elements/nc.html#mayContain_class_tab

Elements

- `<syllable>`
 - One `<syl>` child
 - One or more `<neume>` children
 - neume component `<nc>` children, these are the individual notes that form part of the neume
 - `<nc>` can have children too:
 - `<episema>` ○ `<oriscus>`
 - `<hispanTick>` ○ `<quilisma>`
 - `<liquescent>` ○ `<strophicus>`

```
<neume>  
  <nc tilt="e"/>  
  <nc>  
    <oriscus/>  
  </nc>  
  <nc tilt="ne" intm="u"/>  
</neume>
```



St. Gall

https://music-encoding.org/guidelines/v5/elements/nc.html#mayContain_class_tab

Elements

- `<syllable>`
 - One `<syl>` child
 - One or more `<neume>` children
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 - `<nc>` can have children too:
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 - `<liquescent>` ○ `<strophicus>`

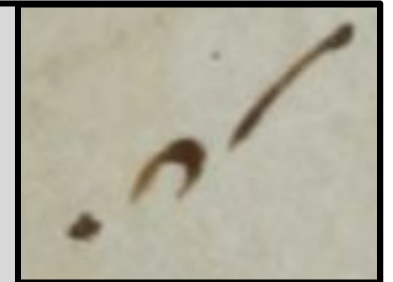
Attributes for `<episema>`:

- ◆ `@form` 'v' (vertical) or 'h' (horizontal)
- ◆ `@place` placement on `<nc>`

Attributes for `<liquescent>`:

- ◆ `@curve` direction of curve
- ◆ `@looped` boolean (when present, 'true')

```
<neume>
  <nc tilt="e"/>
  <nc>
    <oriscus/>
  </nc>
  <nc tilt="ne" intm="u"/>
</neume>
```



St. Gall

https://music-encoding.org/guidelines/v5/elements/nc.html#mayContain_class_tab

Attributes of neume components:

- ❖ **@pname** (pitch name) {[a-g] | unknown}
- ❖ **@oct** (octave number) {[0-9]}
- ❖ **@rellen** (length of pen stroke relative to previous <nc> in the neume) {s = shorter | l = longer}
- ❖ **@tilt** (direction of pen stroke) {n | ne | e | se | s | sw | w | nw}
- ❖ **@con** (connection to the previous <nc> in the neume) {g = gapped | l = looped | e = extended}
- ❖ **@ligated** {true | false}

Attributes of neume components in stafless notation:

- ❖ **@intm** (melodic interval relative to previous <nc> in the neume) {u = up | d = down | s = same}
- ❖ **@s-shape** (described by the direction of the initial penstroke) {n = north | s = south | ...}
- ❖ **@curve** (pen stroke) {c = clockwise | a = anti-clockwise}



```
<neume>  
  <nc tilt="ne"/>  
  <nc tilt="se" relen="s" intm="d"/>  
</neume>
```



```
<neume>  
  <nc tilt="ne"/>  
  <nc tilt="se" relen="l" intm="d"/>  
</neume>
```

Neumatic connections

Neutral – High – Low neumes

<neume>

<nc curve="a"/>

<nc tilt="ne" relen="l" intm="u"/>

<nc tilt="se" relen="s"
intm="d" curve="c"/>

</neume>



<neume>

<nc tilt="se"/>

<nc con="g" tilt="ne" relen="l"
intm="u"/>

<nc tilt="se" relen="l" intm="d"/>

</neume>



<neume>

<nc tilt="se"/>

<nc tilt="ne" relen="l" intm="u"/>

<nc tilt="se" intm="d"/>

</neume>



<neume>

<nc s-shaped="s"/>

<nc con="l" tilt="ne" intm="u"/>

<nc tilt="se" relen="s" intm="d"/>

</neume>



MEI has a long list of <elements> and @attributes employed to capture the meaning of early notations



6 Repertoire: Neume Notation

This chapter describes the elements, model classes, and attribute classes that are part of the MEI.neumes module.

6.1 Overview of the Neumes Module

The MEI Neumes Module represents the community's attempt to create a standardized set of rules that encapsulate in a logical, systematic, and unequivocal way the musical information represented and conveyed by Western European neumatic notations (beginning with the late ninth century and continuing to the printed books of the twentieth). Most neume notation is used to set music to an existing text. The syllable is the fundamental unit of structure, with the neumes themselves serving as a means of "sonifying" the text. A syllable may be expressed via one or more neumes, with the particular neume shape chosen depending on the pitch contour that is being employed and the desired interpretation.

The `syllable` element is used as the primary organizational element for neume notation within a `layer` element. Within `syllable`, the `syl` element defined in the `MEI.shared` module is used for encoding the textual content, while the `neume` and `nc` elements are used to encode the neumes themselves. Within these Neumes Module elements, other standard MEI mechanisms are available to accommodate, for example, editorial or critical markup.

6.1.1 Basic four elements

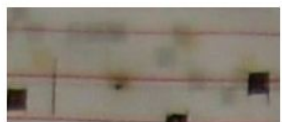
The following four elements are the fundamental components of the Neumes Module:

<> **<syllable>**
Neume notation can be thought of as "neumed text". Therefore, the syllable element provides high-level organization in this repertoire.

Neume notation can be thought of as "neumed text". Therefore, the syllable element provides high-level organization in this repertoire.

<> **<syl>**
Individual lyric syllable.

(syllable) – Individual lyric syllable.



MEI Version: 5.0 (#eb14650)

Guidelines

Modules

Elements

Model Classes

Macro Groups

Attribute Classes

Data Types

6.1 Overview of the Neumes Module

6.1.1 Basic four elements

6.2 Neumes Module Background

6.3 Neume Notation and MEI

6.4 Samples of MEI encodings

6.4.1 Elements

6.4.2 Neume component attributes

6.4.3 Custos attributes

6.4.4 Episema attributes

6.4.5 Liquescent attributes

6.4.6 Old Hispanic tick attributes

6.4.7 Quilisma attribute

6.4.8 Significant letters attribute

6.4.9 Note

6.4.10 Basic Encoding – Syllable

6.4.11 Manuscripts

6.5 Bibliographic References

<https://music-encoding.org/guidelines/v5/content/neumes.html>



WORK IN PROGRESS

lines 5 and 3, respectively.

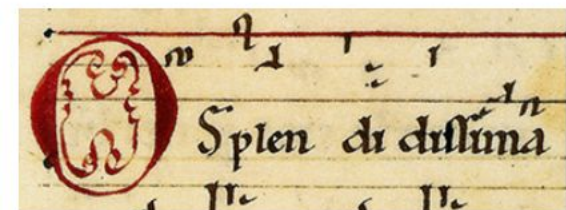


Figure 109.

```
<music meiversion="5.0">
  <body>
    <mdiv>
      <score>
        <section>
          <staff n="1">
            <layer>
              <syllable>
                <syl n="initial">
                  <rend color="red">O</rend>
                </syl>
              <neume>
                <nc oct="3" pname="e"/>
                <nc oct="2" pname="d"/>
                <nc oct="3" pname="e"/>
              </neume>
            </syllable>
            <syllable>
              <syl>splen_</syl>
              <neume>
                <nc oct="3" pname="g"/>
                <nc oct="3" pname="e"/>
              </neume>
              <neume>
                <nc oct="3" pname="d"/>
                <nc oct="3" pname="e"/>
              </neume>
            </syllable>
            <syllable>
              <syl>di_</syl>
              <neume>
                <nc tilt="n" oct="3" pname="f"/>
                <nc tilt="se" con="g" oct="3" pname="d"/>
                <nc tilt="se" con="g" oct="3" pname="c"/>
              </neume>
            </syllable>
          </layer>
        </staff>
      </score>
    </mdiv>
  </body>
</music>
```

MEI Version: 5.0 (#eb14650)

[Guidelines](#)

[Modules](#)

[Elements](#)

[Model Classes](#)

[Macro Groups](#)

[Attribute Classes](#)

[Data Types](#)

6.1 Overview of the Neumes Module

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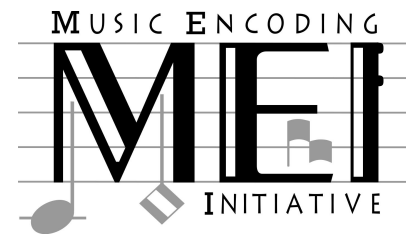
6.4.10 Basic Encoding – Syllable

6.4.11 Manuscripts

6.5 Bibliographic References



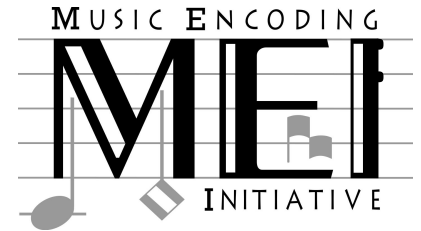
There are **three** main challenges in encoding neumatic notations in



- Convey only partial musical information



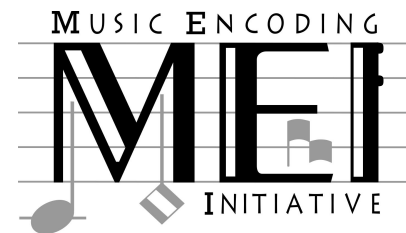
There are **three** main challenges in encoding neumatic notations in



- Convey only partial musical information
- Great variety of signs and existence of different regional styles of early notation
 - Some regional styles occasionally share graphically similar shapes; these are understood by modern scholars to represent the same, a similar or even a *different* musical meaning.



There are **three** main challenges in encoding neumatic notations in



- Convey only partial musical information
- Great variety of signs and existence of different regional styles of early notation
 - Some regional styles occasionally share graphically similar shapes; these are understood by modern scholars to represent the same, a similar or even a *different* musical meaning.
- The neume shapes could either mirror graphically the musical characteristics of the sound being represented or the meaning attached to them was ruled by conventions



What's the impact of these challenges on music encoding?

In the next slides we'll see some **different neume shapes that are given the same name by medieval scribes.**

We start with the oriscus

Ex. from: J. Grier, *Musical Notation in the West*, Cambridge University Press, 2021, p. 30

SG 359

Laon 239

Char 47

Albi 44

<oriscus>



Neumatic

Scripts:

St Gall

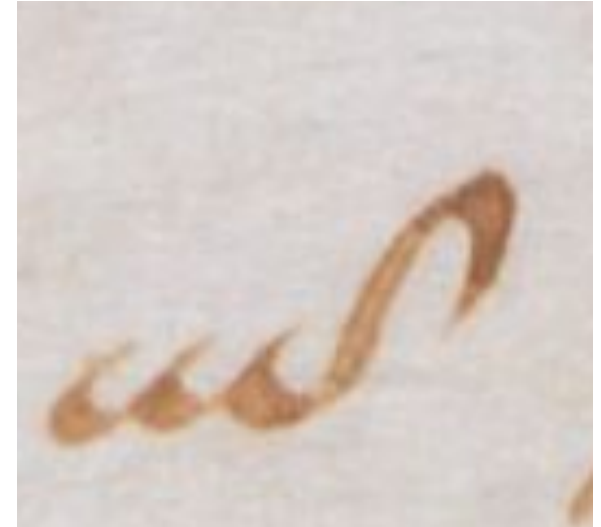
Lotharingian

Breton

Aquitanian

The **quilisma** is a neume found in most early notational dialects, and it gives information on a specific vocal delivery

<quilisma>



It is usually between two notes a third apart. Medieval music theorists spoke of it as a **trembling and rising sound**.

	SG 359	Laon 239	Char 47	Albi 44
<quilisma>				
<oriscus>				
NOTATIONS:	St Gall	Lotharingian	Breton	Aquitanian

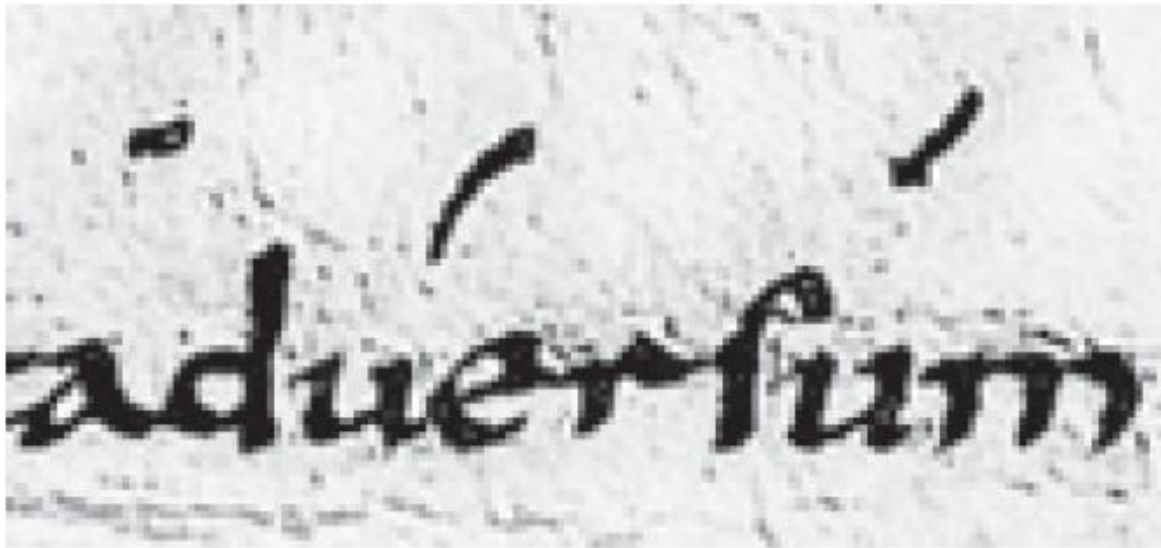
Ex. from: J. Grier, *Musical Notation in the West*, Cambridge University Press, 2021, p. 30

The encoding must be conceived as a coherent system of elements and related attributes that allow us to capture the graphic quality of the neumes and their meaning.

The level of graphical sophistication of early notations



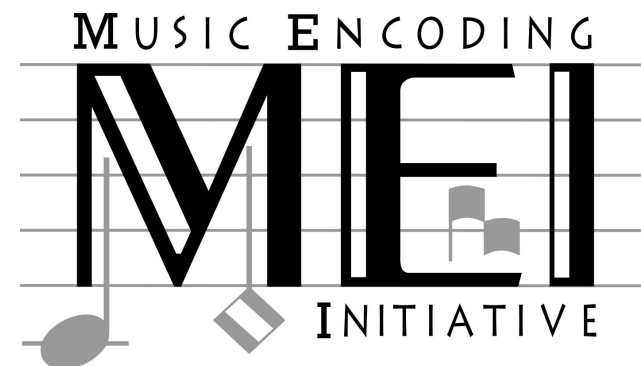
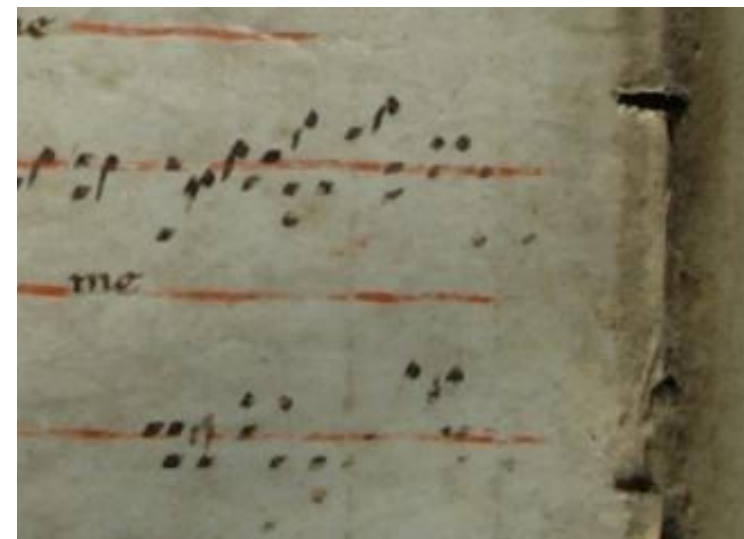
1 2 3



Chartres 47,
fol. 6v,
adversum

Ex. from: J. Grier, *Musical Notation in the West*, Cambridge University Press, 2021, p. 29

Developing MEI for
early music is great fun
for palaeographers!



Developing MEI for early music is great fun for palaeographers!

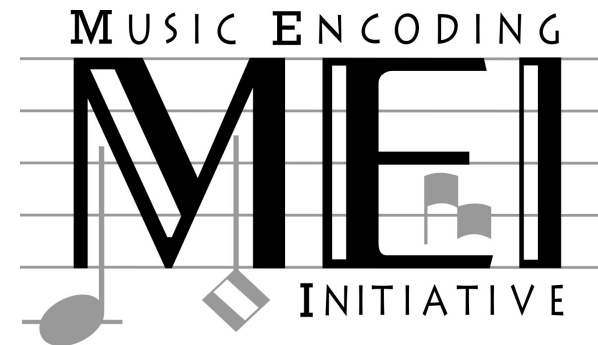


MEI challenges our modern understanding of musical notation and forces us to clarify the difference between what we **see** and what we **know**, or **believe to know**.



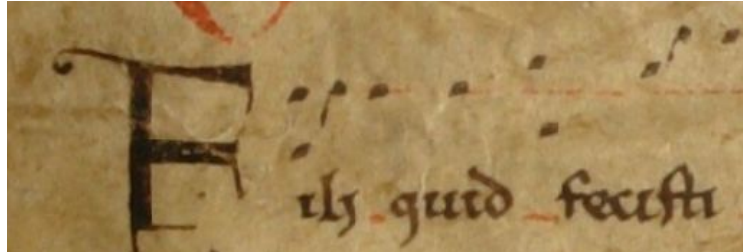
The ongoing **digital revolution** holds the potential to unveil unforeseen possibilities which could challenge and overturn traditional research methodologies and allow entirely new opportunities for music research.

Automatic analysis across larger repertoires would probably allow us to discover new patterns of usage and work out the meaning of early neumatic scripts!





```
<syllable xml:id="m-43558eee-8a6a-11ee-a3f2-3645f29ecd15">
  <syl xml:id="m-43558f20-8a6a-11ee-a3f2-3645f29ecd15">Fi</syl>
  <neume xml:id="m-43558f52-8a6a-11ee-a3f2-3645f29ecd15">
    <nc loc="0" xml:id="m-43558f8e-8a6a-11ee-a3f2-3645f29ecd15"/>
    <nc loc="-2" xml:id="m-43558fc0-8a6a-11ee-a3f2-3645f29ecd15"/>
    <nc loc="0" tilt="ne" xml:id="m-43558ff2-8a6a-11ee-a3f2-3645f29ecd15"/>
  </neume>
</syllable>
<syllable xml:id="m-43559024-8a6a-11ee-a3f2-3645f29ecd15">
  <syl xml:id="m-4355904c-8a6a-11ee-a3f2-3645f29ecd15">li</syl>
  <neume xml:id="m-4355907e-8a6a-11ee-a3f2-3645f29ecd15">
    <nc loc="0" xml:id="m-435590b0-8a6a-11ee-a3f2-3645f29ecd15"/>
  </neume>
</syllable>
<syllable xml:id="m-435590d8-8a6a-11ee-a3f2-3645f29ecd15">
  <syl xml:id="m-4355910a-8a6a-11ee-a3f2-3645f29ecd15">quid</syl>
  <neume xml:id="m-4355913c-8a6a-11ee-a3f2-3645f29ecd15">
    <nc loc="0" xml:id="m-43559178-8a6a-11ee-a3f2-3645f29ecd15"/>
  </neume>
</syllable>
```



Thank you!

